Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *MRBENCH* | ***Context:*** *Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

HIGH priority per elicitation. The annotation pool is four MBZUAI CS post-graduates [Q31] explicitly without required teaching experience [Q32], with no documented South Asian educational expertise. The user expects systematic divergence [elicitation Q4] between Indian teachers' guidance-quality judgments and this annotator pool's, particularly on direct correction vs. Socratic probing, exam-readiness framing, and teacher-directed pedagogy. Field evidence corroborates: Indian teachers structurally value direct, curriculum-aligned outputs over Western-default Socratic scaffolding [WEB-15]. Inter-annotator agreement (Cohen's κ=0.71, Fleiss' κ=0.65) [Q39, Q61] demonstrates internal consistency within the MBZUAI pool but provides no evidence of convergent validity with Indian teachers — the gold standard [Q68] reflects MBZUAI annotators' implicit pedagogical model. LLM-critic correlations were predominantly negative [Q107], confirming pedagogical judgments are not easily automatable. The most-critical dimension's labels were calibrated by annotators whose pedagogical priors likely diverge from Indian professional teachers'. No India-specific annotation study exists [WEB-19].

ID	Evidence
Q31	“The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English.” (p.5)
Q32	“We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient.” (p.5)
Q4	“General domain-agnostic natural language generation (NLG) metrics (Lin, 2004; Popovic´, 2017; Post, 2018; Gao et al., 2020; Liu et al., 2023) are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions.” (p.1)
Q39	“To measure inter-annotator agreement, we computed Fleiss' kappa value, which for this annotation experiment equals 0.65, indicating substantial agreement.” (p.5)
Q61	“The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012).” (p.6)
Q68	“We consider human-based evaluations as gold standard.” (p.7)
Q107	“Across both LLMs, it can be observed that most of the correlation scores are negative (except for the human-likeness dimension), indicating that the annotations from the LLMs are unreliable for the challenging pedagogical dimensions.” (p.8)
WEB-15	Indian teachers face structural misalignment with Western-default AI outputs; value curriculum-aligned, direct guidance (https://arxiv.org/html/2507.00456v1)
WEB-19	No India-specific or cross-cultural ITS evaluation framework cited in 2025 survey (https://arxiv.org/html/2510.22581v1)

Strengths

- Rigorous in-house training and validation protocols [Q33, Q34, Q35, Q36, Q135, Q136, Q137]
- Substantial inter-annotator agreement within the MBZUAI pool (Cohen's κ=0.71) [Q61]
- Annotator demographics are transparently documented [Q31, Q6, Q32], enabling regional gap analysis
- Limitations on LLM-as-evaluator are explicitly reported [Q107, Q114]

ID	Evidence

Q33	“To control the annotation workflow and ensure quality, we opted not to use public annotation outsourcing platforms such as Prolific or MTurk, which allowed us to implement rigorous training protocols and a robust validation mechanism for the annotations.” (p.5)
Q34	“First, we provided all annotators with comprehensive training, including an interactive training document (see Section C for more details) and oral instructions.” (p.5)
Q35	“Following this, we conducted validation pilot study to evaluate the annotation quality and the annotators' understanding of the instructions before rolling out the large-scale human evaluation detailed in Section 5.2.” (p.5)
Q36	“In this validation pilot study, all four annotators iteratively reviewed the annotation scheme and guidelines.” (p.5)
Q135	“As discussed in Section 5.2, prior to commencing large-scale human annotation, we implemented a two-phase interactive training and evaluation protocol and asked each annotator to undertake training.” (p.13)
Q136	“Subsequently, we assessed annotators' understanding through a structured quiz, as is shown in a screenshot presented in Figure 5.” (p.13)
Q137	“Additionally, we developed a comprehensive set of annotation guidelines, serving as a reference for annotators during the large-scale annotation process.” (p.13)
Q61	“The annotators reached an average Cohen's kappa score of 0.71, which indicates substantial inter-annotator agreement (McHugh, 2012).” (p.6)
Q31	“The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English.” (p.5)
Q6	“Kaushal Kumar Maurya, KV Aditya Srivatsa, Ksenia Petukhova and Ekaterina Kochmar. Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE.” (p.1)
Q32	“We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient.” (p.5)
Q107	“Across both LLMs, it can be observed that most of the correlation scores are negative (except for the human-likeness dimension), indicating that the annotations from the LLMs are unreliable for the challenging pedagogical dimensions.” (p.8)
Q114	“We also assess the feasibility of LLMs as evaluators in this context by correlating their judgements with human annotations, indicating that they are often unreliable.” (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels do not reflect Indian regional stakeholder perspectives. Annotators are MBZUAI CS post-graduates [Q31] with no required teaching experience [Q32] and no documented South Asian pedagogical expertise.

OC-2: Systematic disagreement is expected per user elicitation (Q4) and field evidence [WEB-15]. Indian teachers prioritize direct correction, exam-readiness, and teacher-directed pedagogy; MBZUAI annotators were instructed to judge from a 'student or potential AI tutor user' perspective [Q32], not a teacher's.

OC-3: Annotator demographics are documented [Q31, Q6]: four annotators, two male / two female, all post-graduate CS, MBZUAI Abu Dhabi. No teaching experience required [Q32]. No information on cultural/national background beyond institutional affiliation.

OC-4: Re-annotation by representative Indian Grade 1–8 teachers is strongly recommended for this deployment, particularly on the 'providing guidance' dimension. No such re-annotation exists [WEB-19].

OC-5: Aggregation uses pairwise inter-annotator agreement on overlapping subset (40 of 192 instances) [Q57]; the remaining 152 instances are single-annotated. With a homogeneous annotator pool [Q31], minority pedagogical perspectives (e.g., direct-correction-favoring) cannot be detected.

OC-6: Convergent validity concern: gold labels [Q68] correlate with MBZUAI annotators' Western/generic-academic pedagogical priors, not Indian teachers' judgments. External validity concern: original judgments are unlikely to generalize to the Indian metro teacher target context (elicitation Q4, [WEB-15]).

ID	Evidence
Q31	“The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English.” (p.5)
Q32	“We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient.” (p.5)
Q6	“Kaushal Kumar Maurya, KV Aditya Srivatsa, Kseniia Petukhova and Ekaterina Kochmar. Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE.” (p.1)
WEB-15	Indian teachers face structural misalignment with Western-default AI outputs; value curriculum-aligned, direct guidance (https://arxiv.org/html/2507.00456v1)
Q4	“General domain-agnostic natural language generation (NLG) metrics (Lin, 2004; Popovic´, 2017; Post, 2018; Gao et al., 2020; Liu et al., 2023) are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions.” (p.1)
WEB-19	No India-specific or cross-cultural ITS evaluation framework cited in 2025 survey (https://arxiv.org/html/2510.22581v1)
Q57	“A total of 192 instances were annotated, with 40 of those annotated independently by two annotators (10 instances from Bridge and 30 from MathDial) allowing us to calculate pairwise inter-annotator agreement.” (p.6)
Q68	“We consider human-based evaluations as gold standard.” (p.7)

Information Gaps

- No measurement of inter-annotator agreement among Indian professional teachers on the same MRBench instances
- No published cross-cultural comparison of pedagogical-quality judgments on AI tutor responses
- No information on the annotators' national/cultural backgrounds beyond institutional affiliation

Requires Expert Verification

- Pilot re-annotation of a MRBench subset by Delhi/Mumbai Grade 1–8 mathematics teachers, with κ comparison to original labels
- Targeted review of the 'providing guidance' dimension's desired labels by CBSE/NCERT-experienced educators

---

Remediation

Gap 1: Gold-standard labels were produced by MBZUAI CS post-graduates without required teaching experience [Q31, Q32] and no documented South Asian pedagogical expertise; user expects systematic divergence on guidance-quality judgments [elicitation Q4].

Recommendation: Conduct re-annotation of a representative MRBench subset (e.g., 40–60 instances spanning Bridge and MathDial) by 3–4 Delhi/Mumbai Grade 1–8 mathematics teachers with CBSE/ICSE/State Board representation. Compute Cohen's κ between Indian teachers and original MBZUAI labels per dimension; treat per-dimension divergence (especially on 'providing guidance') as the deployment-relevant validity adjustment.

Gap 2: 152 of 192 MRBench instances are single-annotated [Q57], so minority pedagogical perspectives within an already-homogeneous annotator pool [Q31] cannot be detected.

Recommendation: When recruiting Indian teacher re-annotators, ensure full overlap on the re-annotated subset (every instance labeled by ≥2 teachers) and report per-instance disagreement, not just aggregate κ. This surfaces minority-perspective patterns relevant to the diverse Indian metro teaching workforce.

ID	Evidence

Q31	“The annotation team consisted of two male and two female annotators, with all four annotators holding at least a post-graduate degree in Computer Science and being proficient in English.” (p.5)
Q32	“We note that for this study, we do not require annotators to have direct teaching experience, as understanding of the mathematical tasks at the middle school level and being able to judge the responses from the perspective of a potential user of such AI tutors (or a student), rather than specifically a teacher, is sufficient.” (p.5)
Q4	“General domain-agnostic natural language generation (NLG) metrics (Lin, 2004; Popovic´, 2017; Post, 2018; Gao et al., 2020; Liu et al., 2023) are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions.” (p.1)
Q57	“A total of 192 instances were annotated, with 40 of those annotated independently by two annotators (10 instances from Bridge and 30 from MathDial) allowing us to calculate pairwise inter-annotator agreement.” (p.6)